

DEEPTHI DINESHKUMAR

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TECHNICAL SKILLS

Genomics & NGS: RNA-seq, single-cell RNA-seq (Seurat v5, CellChat, scDbfFinder), WES/WGS, variant calling & annotation (GATK, BWA, STAR, Salmon, FASTQC, Prokka, Snippy, PredictSNP2), structural variant analysis, spatial biology exposure

Statistical & Functional Analysis: differential expression (DESeq2), pathway/functional enrichment (clusterProfiler, EnrichR), statistical & machine-learning modeling, multi-omic risk prediction, model validation on independent cohorts

Structural & Molecular Biology: protein structure prediction (AlphaFold2), protein-protein docking (ClusPro), RNA-FM (PyTorch), molecular cloning, protein expression/purification, cell culture

Programming & Tools: Python (Pandas, NumPy, BioPython), R/Bioconductor (ggplot2), Bash, SQL, CLI development (argparse), Git/GitHub, Pytest (95% coverage)

Pipelines, HPC & Databases: Nextflow/nf-core, SLURM/LSF HPC clusters, Docker/Apptainer, GEO, Ensembl, NCBI/SRA, reproducible & auditable workflow design

RESEARCH EXPERIENCE

Bioinformatics Analyst — NGS Analysis & Immunogenomics

Dec 2024 – Apr 2025

Department of Immunology, Madurai Kamaraj University, India

- Independently analyzed whole-exome sequencing data from 150+ patient samples to identify TLR gene polymorphisms and predict functional variant effects on HIV-1 disease progression.
- Developed and validated an automated Python/Nextflow variant-calling and annotation pipeline (GATK, BWA, FASTQC, SAMtools), reducing analysis turnaround time by 30% and improving reproducibility of downstream reports.
- Modeled wild-type and mutant protein structures with AlphaFold2 and performed protein-protein docking (ClusPro) to assess binding-interface consequences of genomic variants relevant to immune recognition.
- Led interpretation of a functionally significant TLR10 missense mutation, translating computational findings into testable biological hypotheses; collaborated with immunologists and presented results to interdisciplinary stakeholders.

RESEARCH PROJECTS

RNA-seq Differential Expression Pipeline — Colorectal Cancer (GEO: GSE50760)

Aug 2026 – Sep 2026

- Built and ran a Nextflow (nf-core/rnaseq) processing pipeline on HPC (SLURM, Apptainer containers), aligning Illumina-sequenced, patient-matched tumor/normal colorectal samples to GRCh38 with STAR/Salmon.
- Independently verified sample identity and patient pairing against primary GEO/SRA metadata (Python, XML parsing) prior to finalizing the experimental design, rather than assuming public dataset labels were correct.
- Performed paired-design DESeq2 differential expression analysis and GO Biological Process enrichment (clusterProfiler), identifying 2,205 significant genes ($\text{padj} < 0.05$, $|\log_2\text{FC}| > 1$) and 489 enriched pathways reflecting a proliferation; extracellular-matrix-remodeling; immune-infiltration signature.
- Diagnosed and resolved a gene-identifier mismatch (symbol vs. Ensembl ID) silently invalidating pathway enrichment output; produced PCA and volcano plot visualizations and a version-controlled, fully documented public portfolio repository (Git/GitHub).

Single-Cell Transcriptome Analysis — Mouse Skin Wound Healing

Jan 2026 – Apr 2026

- Processed multiplexed 10x Genomics single-cell RNA-seq data (Seurat v5): QC, doublet removal (scDbfFinder), PCA, and UMAP clustering, resolving 7 distinct cell populations across spatiotemporal wound-healing conditions.
- Integrated multi-batch datasets using RPCA and sub-clustered fibroblasts into 3 functional states via module scoring with curated phase-specific gene signatures.
- Modeled cell-cell communication with CellChat, uncovering fibroblast-driven signaling trends across the healing time course; version-controlled with Git.

Human Genome Annotation Pipeline — NCI CCDS & Ensembl Integration

Oct 2025 – Dec 2025

- Built a modular Python pipeline integrating NCI CCDS, Ensembl gene annotations, and FASTA sequence data across 30,000+ human protein-coding genes to support large-scale NGS analysis on HPC infrastructure.
- Implemented reproducible, well-documented code with a CLI (argparse), codon usage analysis, and protein translation logic, achieving 95% unit test coverage with Pytest; version-controlled on GitHub.

CRISPR Off-Target Prediction Using RNA-FM Language Model — Project Lead

Sep 2025 – Dec 2025

- Reproduced, validated, and verified a machine-learning CRISPR/Cas9 off-target prediction model (RNA-FM, PyTorch), achieving AUC 0.667 on an imbalanced independent test dataset.
- Thoroughly documented methodology and published a version-controlled, public code repository (Git/GitHub) for reproducibility.

EDUCATION

M.S. Bioinformatics, GPA 3.93/4.0 — Northeastern University, Boston, MA

2025 – 2027

B.Tech Biotechnology, GPA 3.63/4.0 — PSG College of Technology, India

2021 – 2025